

Week 02: Inductive types and recursion

Functional Programming in Lean

Learning goals

- ▶ inductive constructors
- ▶ structure fields
- ▶ Structural recursion on List

Core concepts

- ▶ Suit: alternatives defined by constructors
- ▶ Card: a structure with two fields
- ▶ List cases: `[]` and `x :: xs`

Lean example

```
inductive Suit where
  | clubs | diamonds | hearts | spades

def isRed : Suit → Bool
  | .hearts | .diamonds => true
  | .clubs | .spades => false
```

Lean details

- ▶ Pattern matching on every constructor
- ▶ Recursive call on the list tail
- ▶ Termination from structural recursion

Live coding

- ▶ `isRed`: cases for each suit
- ▶ `countRed`: empty and cons cases
- ▶ Evaluation on a short card list

Check

Base case of countRed

Exercise

- ▶ `countSuit` on a card list
- ▶ Termination argument